

## Qualification Pack



# Welding Technician - Oil & Gas/Shipbuilding/Railways

Options: Perform AR-guided Oil & Gas Welding Scenario & Performance Evaluation/ Perform AR-guided SMAW/GMAW in Railways/ Perform AR-guided SMAW/GMAW in Shipbuilding

QP Code: CSC/Q0214

Version: 1.0

NSQF Level: 4



## Qualification Pack

Capital Goods & Strategic Skill Council || 1st Floor, L-29, Outer Circle, Connaught Place  
New Delhi - 110001 || email:technicaladvisors@cgsc.in

## Qualification Pack

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## Qualification Pack

### CSC/Q0214: Welding Technician - Oil & Gas/Shipbuilding/Railways

#### Brief Job Description

A Welding Technician is responsible for executing welding operations in various industries such as Oil & Gas, Shipbuilding & Railways, utilizing augmented reality (AR) technology to enhance precision, efficiency, and safety. This role involves performing high-quality welding on a variety of materials and components while adhering to industry standards and regulations.

#### Personal Attributes

A strong understanding of welding techniques, materials, and Oil & Gas Industry processes is essential. Familiarity with AR technologies and their applications in manufacturing can further enhance performance. Proficient in using augmented reality tools, welding equipment, and related technology to enhance productivity and accuracy.

#### Applicable National Occupational Standards (NOS)

##### Compulsory NOS:

1. [CSC/N1335: Follow the health and safety practices at the work](#)
2. [CSC/N1336: Coordinate with co-workers to achieve work efficiency](#)
3. [CSC/N0204: Manually weld carbon and low alloy steels by using Metal Arc Welding \(MMAW\)/Shielded Metal Arc Welding \(SMAW\)](#)
4. [CSC/N0209: Manually weld metals by using MIG/MAG welding](#)
5. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

##### Options(Not mandatory):

##### Option 1: Perform AR-guided Oil & Gas Welding Scenario & Performance Evaluation

This toolkit should leverage AR to overlay instructions, measurements and visual cues onto the real world welding environments , improving accuracy & safety.

1. [CSC/N0215: Perform AR-guided SMAW/GMAW in Oil & Gas Industry](#)

##### Option 2: Perform AR-guided SMAW/GMAW in Railways

An AR welding simulator provides a virtual, interactive training experience that enhances learning & skill development in welding. It overlays digital welding techniques onto real world environment allowing learners to practice and refine their skills in a safe controlled setting.

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### 1. [CSC/N0216: Perform AR-guided SMAW/GMAW in Railways](#)

#### Option 3: Perform AR-guided SMAW/GMAW in Shipbuilding

An AR welding simulator provides a virtual, interactive training experience that enhances learning & skill development in welding. It overlays digital welding techniques onto real world environment allowing learners to practice and refine their skills in a safe controlled setting. An AR based toolkit for SMAW/GMAW welding in shipbuilding should include tools and accessories for both welding processes, including power sources, welding guns/electrodes, safety equipment and inspection finishing tools all integrated with an AR application.

### 1. [CSC/N0217: Perform AR-guided SMAW/GMAW in Shipbuilding](#)

## Qualification Pack (QP) Parameters

|                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sector</b>                                             | Capital Goods                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Sub-Sector</b>                                         | Process Plant Machinery, Strategic Manufacturing                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Occupation</b>                                         | Welding and Cutting                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Country</b>                                            | India                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>NSQF Level</b>                                         | 4                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Credits</b>                                            | 4                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Aligned to NCO/ISCO/ISIC Code</b>                      | NCO-2015/7212.0302                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Minimum Educational Qualification &amp; Experience</b> | 12th Class (or equivalent)<br>OR<br>Completed 2nd year of the 3-year diploma after 10 (or equivalent)<br>OR<br>10th grade pass with 3 Years of experience relevant<br>OR<br>11th grade pass with 1.5 years of experience relevant<br>OR<br>Previous relevant Qualification of NSQF Level (3.5) with 1.5 years of experience relevant<br>OR<br>Previous relevant Qualification of NSQF Level (3) with 3 Years of experience relevant |
| <b>Minimum Level of Education for Training in School</b>  |                                                                                                                                                                                                                                                                                                                                                                                                                                     |

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|                                          |                             |
|------------------------------------------|-----------------------------|
| <b>Pre-Requisite License or Training</b> | NA                          |
| <b>Minimum Job Entry Age</b>             | 18 Years                    |
| <b>Last Reviewed On</b>                  | NA                          |
| <b>Next Review Date</b>                  | 08/05/2028                  |
| <b>NSQC Approval Date</b>                | 08/05/2025                  |
| <b>Version</b>                           | 1.0                         |
| <b>Reference code on NQR</b>             | QG-04-IT-04205-2025-V1-CGSC |
| <b>NQR Version</b>                       | 1.0                         |

## Qualification Pack

### CSC/N1335: Follow the health and safety practices at the work

#### Description

This OS unit is about following the appropriate health and safety practices at work. It covers responsibilities towards self and others to ensure a safe work environment.

#### Scope

The scope covers the following :

- Maintain personal health and safety
- Assist in hazard management
- Check the first aid box, firefighting and safety equipment
- Assist in waste management
- Follow the fire safety guidelines
- Follow the emergency and first-aid procedures
- Carry out relevant documentation and review

#### Elements and Performance Criteria

##### *Maintain personal health and safety*

To be competent, the user/individual on the job must be able to:

- PC1.** follow the recommended practices to ensure protection from infections and transmission to others, such as the use of hand sanitiser and face mask
- PC2.** check the work conditions, assess the potential health and safety risks, and take appropriate measures to mitigate them
- PC3.** select and use the appropriate Personal Protective Equipment (PPE) relevant to the task and work conditions
- PC4.** follow the recommended techniques while lifting and moving heavy objects to avoid injury
- PC5.** follow the manufacturer's instructions and workplace safety guidelines while working on heavy machinery, tools and equipment

##### *Assist in hazard management*

To be competent, the user/individual on the job must be able to:

- PC6.** identify existing and potential hazards at work
- PC7.** assess the potential risks and injuries associated with the identified hazards
- PC8.** coordinate with the supervisor or other relevant personnel to prevent or minimise the identified hazards
- PC9.** handle hazardous materials safely and store them in the designated storage

##### *Check the first aid box, firefighting and safety equipment*

To be competent, the user/individual on the job must be able to:

- PC10.** check the first aid box to ensure it is updated with the relevant first aid supplies
- PC11.** check and test the firefighting and various safety equipment to ensure they are in usable condition

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**PC12.** coordinate with the supervisor for the repair and replacement of firefighting and safety equipment

### *Assist in waste management*

To be competent, the user/individual on the job must be able to:

**PC13.** segregate waste into appropriate categories

**PC14.** recycle the recyclable waste appropriately

**PC15.** dispose of the non-recyclable waste in an environment-friendly manner, complying with the applicable regulations

### *Follow the fire safety guidelines*

To be competent, the user/individual on the job must be able to:

**PC16.** use the appropriate type of fire extinguisher to extinguish different types of fires safely

**PC17.** follow the recommended practices for a safe rescue during a fire emergency

**PC18.** coordinate with the fire department to request assistance to extinguish a serious fire

### *Follow the emergency and first-aid procedures*

To be competent, the user/individual on the job must be able to:

**PC19.** follow the organisational health and safety guidelines during workplace emergencies to ensure own and co-workers' safety

**PC20.** follow the recommended practices to minimise loss to organisational property during an emergency

**PC21.** follow the recommended procedure to free a person from electrocution

**PC22.** administer appropriate first aid to the injured personnel

**PC23.** perform Cardiopulmonary Resuscitation (CPR) on a potential victim of cardiac arrest

**PC24.** coordinate with the emergency services to request medical assistance for seriously injured/ ill personnel requiring professional medical attention or hospitalisation

### *Carry out relevant documentation and review*

To be competent, the user/individual on the job must be able to:

**PC25.** carry out appropriate documentation following a health and safety incident at work, including all the required information

**PC26.** coordinate with the relevant personnel to review health and safety conditions at work regularly or following an incident

**PC27.** assist in implementing appropriate changes to improve the health and safety conditions at work

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** the recommended practices to be followed to ensure protection from infections and transmission to others, such as the use of hand sanitiser and face mask

**KU2.** the importance and process of checking the work conditions, assessing the potential health and safety risks, and take appropriate measures to mitigate them

**KU3.** the importance and process of selecting and using the appropriate PPE relevant to the task and work conditions



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- KU4.** the recommended techniques to be followed while lifting and moving heavy objects to avoid injury
- KU5.** the importance of following the manufacturer's instructions and workplace safety guidelines while working on heavy machinery, tools and equipment
- KU6.** the importance and process of identifying existing and potential hazards at work
- KU7.** the process of assessing the potential risks and injuries associated with the various hazards
- KU8.** how to prevent or minimise different types of hazards
- KU9.** how to handle and store hazardous materials safely
- KU10.** the importance of ensuring the first aid box is updated with the relevant first aid supplies
- KU11.** the process of checking and testing the firefighting and various safety equipment to ensure they are in a usable condition
- KU12.** the criteria for segregating waste into appropriate categories
- KU13.** the appropriate methods for recycling the recyclable waste
- KU14.** the process of disposing of the non-recyclable waste safely and the applicable regulations
- KU15.** Use of different types of fire extinguishers to extinguish different types of fires
- KU16.** the recommended practices to be followed for a safe rescue during a fire emergency
- KU17.** how to request assistance from the fire department to extinguish a serious fire
- KU18.** the appropriate practices to be followed during workplace emergencies to ensure safety and minimise loss to organisational property
- KU19.** common health and safety hazards present in a work environment, associated risks, and how to mitigate them
- KU20.** safe working practices to be followed while working at various hazardous sites and using electrical equipment
- KU21.** the importance of ensuring easy access to firefighting and safety equipment
- KU22.** the appropriate preventative and remedial actions to be taken in the case of exposure to toxic materials, such as poisonous chemicals and gases
- KU23.** various causes of fire in different work environments and the recommended precautions to be taken to prevent fire accidents
- KU24.** different methods of extinguishing fire
- KU25.** different materials used for extinguishing fire, such as sand, water, foam, CO2, dry powder, etc.
- KU26.** the applicable rescue techniques to be followed during a fire emergency
- KU27.** the importance of placing safety signs and instructions at strategic locations in a workplace and following them
- KU28.** different types of first aid treatment to be provided for different types of injuries
- KU29.** potential injuries associated with incorrect manual handling
- KU30.** how to move an injured person safely
- KU31.** various hazards associated with the use of various machinery, tools, implements, equipment and materials
- KU32.** the importance of ensuring no obstruction and free access to fire exits
- KU33.** how to free a person from electrocution safely
- KU34.** how to administer appropriate first aid to an injured person

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- KU35.** how to perform Cardiopulmonary Resuscitation (CPR)
- KU36.** the importance of coordinating with the emergency services to request urgent medical assistance for persons requiring professional medical attention or hospitalisation
- KU37.** the appropriate documentation to be carried out following a health and safety incident at work, and the relevant information to be included
- KU38.** the importance and process of reviewing the health and safety conditions at work regularly or following an incident
- KU39.** the importance and process of implementing appropriate changes to improve the health and safety conditions at work

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work-related notes and records
- GS2.** communicate clearly and politely with co-workers and clients
- GS3.** read the relevant literature to get the latest updates about the field of work
- GS4.** listen attentively to understand the information being shared
- GS5.** plan and prioritise tasks to ensure timely completion
- GS6.** take quick decisions to deal with workplace emergencies and accidents
- GS7.** identify possible disruptions to work and take appropriate preventive measures
- GS8.** coordinate with the co-workers to achieve the work objectives
- GS9.** evaluate all possible solutions to a problem to select the best one

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                              | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Maintain personal health and safety</i>                                                                                                                    | <b>7</b>     | <b>12</b>       | -             | -          |
| <b>PC1.</b> follow the recommended practices to ensure protection from infections and transmission to others, such as the use of hand sanitiser and face mask | 2            | 3               | -             | -          |
| <b>PC2.</b> check the work conditions, assess the potential health and safety risks, and take appropriate measures to mitigate them                           | 1            | 2               | -             | -          |
| <b>PC3.</b> select and use the appropriate Personal Protective Equipment (PPE) relevant to the task and work conditions                                       | 1            | 2               | -             | -          |
| <b>PC4.</b> follow the recommended techniques while lifting and moving heavy objects to avoid injury                                                          | 1            | 3               | -             | -          |
| <b>PC5.</b> follow the manufacturer's instructions and workplace safety guidelines while working on heavy machinery, tools and equipment                      | 2            | 2               | -             | -          |
| <i>Assist in hazard management</i>                                                                                                                            | <b>4</b>     | <b>10</b>       | -             | -          |
| <b>PC6.</b> identify existing and potential hazards at work                                                                                                   | 1            | 1               | -             | -          |
| <b>PC7.</b> assess the potential risks and injuries associated with the identified hazards                                                                    | 1            | 3               | -             | -          |
| <b>PC8.</b> coordinate with the supervisor or other relevant personnel to prevent or minimise the identified hazards                                          | 1            | 3               | -             | -          |
| <b>PC9.</b> handle hazardous materials safely and store them in the designated storage                                                                        | 1            | 3               | -             | -          |
| <i>Check the first aid box, firefighting and safety equipment</i>                                                                                             | <b>3</b>     | <b>7</b>        | -             | -          |
| <b>PC10.</b> check the first aid box to ensure it is updated with the relevant first aid supplies                                                             | 1            | 2               | -             | -          |
| <b>PC11.</b> check and test the firefighting and various safety equipment to ensure they are in usable condition                                              | 1            | 3               | -             | -          |

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| Assessment Criteria for Outcomes                                                                                                                                                   | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC12.</b> coordinate with the supervisor for the repair and replacement of firefighting and safety equipment                                                                    | 1            | 2               | -             | -          |
| <i>Assist in waste management</i>                                                                                                                                                  | <b>3</b>     | <b>8</b>        | -             | -          |
| <b>PC13.</b> segregate waste into appropriate categories                                                                                                                           | 1            | 3               | -             | -          |
| <b>PC14.</b> recycle the recyclable waste appropriately                                                                                                                            | 1            | 3               | -             | -          |
| <b>PC15.</b> dispose of the non-recyclable waste in an environment-friendly manner, complying with the applicable regulations                                                      | 1            | 2               | -             | -          |
| <i>Follow the fire safety guidelines</i>                                                                                                                                           | <b>3</b>     | <b>12</b>       | -             | -          |
| <b>PC16.</b> use the appropriate type of fire extinguisher to extinguish different types of fires safely                                                                           | 1            | 4               | -             | -          |
| <b>PC17.</b> follow the recommended practices for a safe rescue during a fire emergency                                                                                            | 1            | 4               | -             | -          |
| <b>PC18.</b> coordinate with the fire department to request assistance to extinguish a serious fire                                                                                | 1            | 4               | -             | -          |
| <i>Follow the emergency and first-aid procedures</i>                                                                                                                               | <b>7</b>     | <b>12</b>       | -             | -          |
| <b>PC19.</b> follow the organisational health and safety guidelines during workplace emergencies to ensure own and co-workers' safety                                              | 1            | 2               | -             | -          |
| <b>PC20.</b> follow the recommended practices to minimise loss to organisational property during an emergency                                                                      | 1            | 3               | -             | -          |
| <b>PC21.</b> follow the recommended procedure to free a person from electrocution                                                                                                  | 1            | 2               | -             | -          |
| <b>PC22.</b> administer appropriate first aid to the injured personnel                                                                                                             | 1            | 2               | -             | -          |
| <b>PC23.</b> perform Cardiopulmonary Resuscitation (CPR) on a potential victim of cardiac arrest                                                                                   | 1            | 2               | -             | -          |
| <b>PC24.</b> coordinate with the emergency services to request medical assistance for seriously injured/ ill personnel requiring professional medical attention or hospitalisation | 2            | 1               | -             | -          |

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| Assessment Criteria for Outcomes                                                                                                        | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Carry out relevant documentation and review</i>                                                                                      | <b>3</b>     | <b>9</b>        | -             | -          |
| <b>PC25.</b> carry out appropriate documentation following a health and safety incident at work, including all the required information | 1            | 3               | -             | -          |
| <b>PC26.</b> coordinate with the relevant personnel to review health and safety conditions at work regularly or following an incident   | 1            | 3               | -             | -          |
| <b>PC27.</b> assist in implementing appropriate changes to improve the health and safety conditions at work                             | 1            | 3               | -             | -          |
| <b>NOS Total</b>                                                                                                                        | <b>30</b>    | <b>70</b>       | -             | -          |

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### National Occupational Standards (NOS) Parameters

|                            |                                                                                                                                                                                                  |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NOS Code</b>            | CSC/N1335                                                                                                                                                                                        |
| <b>NOS Name</b>            | Follow the health and safety practices at the work                                                                                                                                               |
| <b>Sector</b>              | Capital Goods                                                                                                                                                                                    |
| <b>Sub-Sector</b>          | Machine Tools, Process Plant Machinery, Dies, Moulds and Press Tools, Electrical and Power Machinery, Plastics Manufacturing Machinery, Light Engineering Goods, Textile Manufacturing Machinery |
| <b>Occupation</b>          | Machining                                                                                                                                                                                        |
| <b>NSQF Level</b>          | 3                                                                                                                                                                                                |
| <b>Credits</b>             | TBD                                                                                                                                                                                              |
| <b>Version</b>             | 2.0                                                                                                                                                                                              |
| <b>Last Reviewed Date</b>  | 01/10/2025                                                                                                                                                                                       |
| <b>Next Review Date</b>    | 01/10/2028                                                                                                                                                                                       |
| <b>NSQC Clearance Date</b> | 01/10/2025                                                                                                                                                                                       |

## Qualification Pack

### CSC/N1336: Coordinate with co-workers to achieve work efficiency

#### Description

This OS unit is about working in coordination with co-workers to achieve the work objectives efficiently. It also covers practising inclusion at work.

#### Scope

The scope covers the following :

- Work effectively with co-workers
- Communicate effectively with co-workers
- Practice inclusion at work

#### Elements and Performance Criteria

##### *Work effectively with co-workers*

To be competent, the user/individual on the job must be able to:

- PC1.** plan daily tasks at work to ensure their timely completion and efficient use of time
- PC2.** carry out work responsibilities adhering to the limits of authority
- PC3.** follow the supervisor's instructions to ensure adherence to the applicable quality standards and timescales
- PC4.** coordinate with the co-workers to achieve the work objectives efficiently
- PC5.** prepare the relevant documents and reports as per the supervisor's instructions, providing appropriate information clearly and systematically
- PC6.** coordinate with the supervisor or relevant personnel to deal with out of authority tasks and concerns
- PC7.** mentor and assist subordinates in the execution of their work responsibilities
- PC8.** identify possible disruptions to work through coordination with the relevant stakeholders and take appropriate preventive measures
- PC9.** use various resources efficiently to ensure maximum utilisation and minimum wastage
- PC10.** follow the recommended practices to avoid and resolve conflicts at work
- PC11.** follow the relevant organisational policies to ensure disciplined behaviour with maximum productivity at work

##### *Communicate effectively with co-workers*

To be competent, the user/individual on the job must be able to:

- PC12.** follow the organisational policy for the efficient and timely dissemination of information to the authorised personnel
- PC13.** communicate clearly and politely to ensure effective communication with co-workers
- PC14.** follow the appropriate techniques for active listening during interactions

##### *Practice inclusion at work*

To be competent, the user/individual on the job must be able to:

- PC15.** empathise with Persons with Disabilities (PwD)

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**PC16.** adopt gender-neutral behaviour at work

### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the importance and process of effective communication in the workplace
- KU2.** the barriers to effective communication and how to overcome them
- KU3.** the importance of teamwork in an organisation's and individual's success
- KU4.** the importance of active listening in the work environment
- KU5.** the appropriate techniques to be followed for active listening
- KU6.** importance of avoiding casual expletives and unpleasant terms while communicating professional circles
- KU7.** the importance of maintaining discipline and ethical behaviour at work
- KU8.** the common reasons for interpersonal conflict and how to resolve them
- KU9.** the importance of developing effective work relationships for professional success
- KU10.** how expressing and addressing grievances appropriately and effectively
- KU11.** the importance and process of planning daily tasks to ensure their timely completion and efficient use of time
- KU12.** the importance of adhering to the limits of authority at work
- KU13.** the importance of following the applicable quality standards and timescales at work
- KU14.** the importance of coordinating with the co-workers to achieve the work objectives efficiently
- KU15.** the relevant documentation requirements
- KU16.** the importance of providing appropriate information clearly and systematically in work documents
- KU17.** the escalation matrix to be followed to deal with out of authority tasks and concerns
- KU18.** the importance and process of mentoring and assisting subordinates in the execution of their work responsibilities
- KU19.** how to identify possible disruptions to work prevent them
- KU20.** how to use various resources efficiently to ensure maximum utilisation and minimum wastage
- KU21.** the recommended practices to be followed at work to avoid and resolve conflicts at work
- KU22.** the importance and process of efficient and timely dissemination of information to the authorised personnel
- KU23.** how to communicate clearly and politely to ensure effective communication
- KU24.** the importance of following the recommended practices to ensure an inclusive environment for PwD and all genders at work

### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work-related notes and records



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- GS2.** read work-related and other relevant literature
- GS3.** communicate politely and -professionally
- GS4.** listen attentively to understand the information or instructions being shared
- GS5.** plan and prioritise tasks to ensure timely completion
- GS6.** take prompt decisions to deal with workplace emergencies and accidents
- GS7.** evaluate all possible solutions to a problem to select the best one

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### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                          | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Work effectively with co-workers</i>                                                                                                                   | <b>20</b>    | <b>43</b>       | -             | -          |
| <b>PC1.</b> plan daily tasks at work to ensure their timely completion and efficient use of time                                                          | 2            | 4               | -             | -          |
| <b>PC2.</b> carry out work responsibilities adhering to the limits of authority                                                                           | 2            | 4               | -             | -          |
| <b>PC3.</b> follow the supervisor's instructions to ensure adherence to the applicable quality standards and timescales                                   | 2            | 4               | -             | -          |
| <b>PC4.</b> coordinate with the co-workers to achieve the work objectives efficiently                                                                     | 2            | 4               | -             | -          |
| <b>PC5.</b> prepare the relevant documents and reports as per the supervisor's instructions, providing appropriate information clearly and systematically | 2            | 4               | -             | -          |
| <b>PC6.</b> coordinate with the supervisor or relevant personnel to deal with out of authority tasks and concerns                                         | 2            | 4               | -             | -          |
| <b>PC7.</b> mentor and assist subordinates in the execution of their work responsibilities                                                                | 2            | 4               | -             | -          |
| <b>PC8.</b> identify possible disruptions to work through coordination with the relevant stakeholders and take appropriate preventive measures            | 2            | 4               | -             | -          |
| <b>PC9.</b> use various resources efficiently to ensure maximum utilisation and minimum wastage                                                           | 2            | 4               | -             | -          |
| <b>PC10.</b> follow the recommended practices to avoid and resolve conflicts at work                                                                      | 1            | 4               | -             | -          |
| <b>PC11.</b> follow the relevant organisational policies to ensure disciplined behaviour with maximum productivity at work                                | 1            | 3               | -             | -          |
| <i>Communicate effectively with co-workers</i>                                                                                                            | <b>6</b>     | <b>15</b>       | -             | -          |

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| Assessment Criteria for Outcomes                                                                                                    | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC12.</b> follow the organisational policy for the efficient and timely dissemination of information to the authorised personnel | 2            | 5               | -             | -          |
| <b>PC13.</b> communicate clearly and politely to ensure effective communication with co-workers                                     | 2            | 5               | -             | -          |
| <b>PC14.</b> follow the appropriate techniques for active listening during interactions                                             | 2            | 5               | -             | -          |
| <i>Practice inclusion at work</i>                                                                                                   | <b>4</b>     | <b>12</b>       | -             | -          |
| <b>PC15.</b> empathise with Persons with Disabilities (PwD)                                                                         | 2            | 6               | -             | -          |
| <b>PC16.</b> adopt gender-neutral behaviour at work                                                                                 | 2            | 6               | -             | -          |
| <b>NOS Total</b>                                                                                                                    | <b>30</b>    | <b>70</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                                                                                                                                                  |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NOS Code</b>            | CSC/N1336                                                                                                                                                                                        |
| <b>NOS Name</b>            | Coordinate with co-workers to achieve work efficiency                                                                                                                                            |
| <b>Sector</b>              | Capital Goods                                                                                                                                                                                    |
| <b>Sub-Sector</b>          | Machine Tools, Dies, Moulds and Press Tools, Plastics Manufacturing Machinery, Textile Manufacturing Machinery, Process Plant Machinery, Electrical and Power Machinery, Light Engineering Goods |
| <b>Occupation</b>          | Machining                                                                                                                                                                                        |
| <b>NSQF Level</b>          | 3                                                                                                                                                                                                |
| <b>Credits</b>             | TBD                                                                                                                                                                                              |
| <b>Version</b>             | 2.0                                                                                                                                                                                              |
| <b>Last Reviewed Date</b>  | 01/10/2025                                                                                                                                                                                       |
| <b>Next Review Date</b>    | 01/10/2028                                                                                                                                                                                       |
| <b>NSQC Clearance Date</b> | 01/10/2025                                                                                                                                                                                       |

## Qualification Pack

# CSC/N0204: Manually weld carbon and low alloy steels by using Metal Arc Welding (MMAW)/ Shielded Metal Arc Welding (SMAW)

## Description

This unit is about performing manual metal arc welding (MMAW) for producing various types of joints on carbon and low alloy steels as per the given specifications and standards specified by the organisation.

## Scope

The scope covers the following :

- Prepare for welding operations
- Perform MMAW/SMAW operations
- Perform post-welding operations

## Elements and Performance Criteria

### *Prepare for welding operations*

To be competent, the user/individual on the job must be able to:

- PC1.** identify the work to be done and product specifications by interpreting the product drawing, Welding Procedure Specification (WPS) and job orders
- PC2.** identify the tools, welding machines, measuring instruments, accessories, consumables and input materials (i.e. carbon, low alloy steel etc.) as per the requirements mentioned in WPS or drawing
- PC3.** select and arrange the right material, equipment, fixtures, accessories, welding torch and consumables i.e. electrode, filler wire, shielding gas etc. as per the SOP and job requirements
- PC4.** check the input material, tools and equipment for any defects and that they are as per the required quality standards
- PC5.** prepare the work area for the welding activities
- PC6.** prepare the materials (i.e. plate(1.5 - 24mm)/ sheet (1.5mm)) and joint for welding process
- PC7.** set the MMAW machine and its parameters as per the WPS and SOP
- PC8.** re-dry electrodes as per electrode classification requirement
- PC9.** install the work pieces and fixture on the apparatus and align them with the electrodes as per the job requirements
- PC10.** verify set up by running test weld specimen (scrap plate)

### *Perform MMAW/SMAW operations*

To be competent, the user/individual on the job must be able to:

- PC11.** follow safety precautions during welding work as per SOP and organizational guidelines
- PC12.** start the MMAW machine for welding operations
- PC13.** strike and maintain a stable arc by applying correct technique (i.e. scratch start, tapping techniques) and to avoid welding defects
- PC14.** perform MMAW welding process as per SOP and tack weld the joint at appropriate intervals to produce joints of the specified quality, dimensions and profile

## Qualification Pack

- PC15.** produce fillet and groove joints in 1F/1G, 2F/2G and 3F/ 3G welding positions as specified in WPS by using single or multi-run welds
- PC16.** ensure correct angle of torch, travel speed, direction of weld and feed during the welding operation
- PC17.** maintain proper bead sequence with respect to groove/fillet configurations and positions
- PC18.** monitor the welding process parameters (air pressure, electrode force, electrode distance, gas flow etc.) are within standards by reading the various gauges and correct them if not within standards
- PC19.** measure the final welded piece and compare with the dimensional and geometrical aspects of the weld as prescribed in the WPS and engineering drawing
- PC20.** remove extra material, slag etc. by using brush, chipping hammers, grinders etc., from the welded piece
- PC21.** hammer the work piece to get the desired shape, if there are any welding bulges/distortions
- PC22.** shut down the welding equipment and remove the workpiece after completion of welding activities

### *Perform post-welding operations*

To be competent, the user/individual on the job must be able to:

- PC23.** check the work pieces as per the work instructions for product quality
- PC24.** identify various weld defects by conducting visual inspection, destructive and non-destructive tests on the work pieces
- PC25.** separate the defective pieces which can be repaired/ reworked and pieces which are beyond repair
- PC26.** clean and store all the tools, machine and equipment after completion of work
- PC27.** dispose scrap or waste material in accordance with the company policies and environmental regulations
- PC28.** report to the supervisor about any problems faced or anticipated during the complete process

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant legislation, standards, policies, and procedures followed in the organization
- KU2.** the basic principle of welding process
- KU3.** MMAW welding and its process flow
- KU4.** various types of welding joints (i.e. fillet lap joints, tee fillet joints, corner joints, butt joints (square, single, vee, double vee)) and welding positions (i.e. flat (PA) 1G/1F, horizontal vertical (PB) 2F, horizontal (PC) 2G, vertical upwards (PF) 3F / 3G, vertical downwards (PG) 3F / 3G, Plate to Pipe (Fixed) 5F)
- KU5.** how to read and interpret WPS, welding drawings and symbols
- KU6.**
  - welding specific equipment requirements for MMAW/SMAW welding
  - MMAW equipment: transformers, rectifiers, generators, invertors;
  - Consumables – electrodes, dyes;
  - Welding accessories - holders, cables, welding torch and accessories;
  - Ancillary equipment - power saw, angle, pedestal and straight grinders, tong tester; etc.

## Qualification Pack

- KU7.** SOP recommended by the manufacturer for using tools, measuring instruments, accessories, MMAW welding machine etc. during the welding process
- KU8.** main components and controls of welding equipment
- KU9.** type of current used and implication
- KU10.** ISO colour codes for welding apparatus such as gas cylinder, hoses, electric cables, etc.
- KU11.** joint preparation process: made rust free; cleaned – free from scaling, paint, oil/grease; made dry and free from moisture; edges to be welded prepared as per job requirement - such as flat, square or bevelled; use various machines and techniques for the above (e.g. chamfering machine, grinding and stripping, gas or plasma cutting, etc.); correctly positioned (positioning: devices and techniques; jigs and fixtures; setting up joint in correct position & alignment)
- KU12.** Impact of various welding parameters like voltage, current, gas flow rate, speed, pressure, torch angle, cycle time, electrode distance etc. on the quality and quantity of welding
- KU13.** welding techniques i.e. drag, weave, whip
- KU14.** various materials used for MMAW welding and their properties
- KU15.** SOP recommended by the organisation for operating MMAW welding machine and its accessories
- KU16.** purpose and importance of pre-heating requirements for base metals
- KU17.** • factors that determine weld bead shape  
 • Factors: electrode angles and welding technique (push, perpendicular, drag); arc length; thickness of base metal; travel speed (slow, normal, fast)
- KU18.** • types of beads, characteristics and uses (stringer, weave, weave patterns)  
 • Bead characteristics: spatter deposits, roughness, evenness, fill, crater, overlap
- KU19.** SOP recommended by the organisation for checking irregularities in the product/work piece
- KU20.** • factors that affect weld quality standards  
 • Quality standards: required parameters for dimensional accuracy; weld finishes are built up to the full section of the weld; joins at stop/start positions merge smoothly; weld surface is (free from cracks; substantially free from porosity; free from any pronounced hump or crater; substantially free from shrinkage cavities; substantially free from trapped slag; substantially free from arcing or chipping marks); fillet welds are (equal in leg length, slightly convex in profile (where applicable), size of the fillet equivalent to the thickness of the material welded); weld contour is (of linear and of uniform profile; smooth and free from excessive undulations; regular and has an even ripple formations); welds are adequately fused, and there is minimal undercut, overlap and surface inclusions etc.
- KU21.** • various defects associated with the MMAW/SMAW welding process  
 • Weld defects: lack of continuity of the weld; uneven and irregular ripple formation; excessive spatter; incorrect weld size or profile; burn through; undercutting; overlap; inclusions; distortion; porosity; internal cracks; surface cracks; lack of fusion or incomplete fusion; lack of penetration; excessive penetration; gouges; stray arc strikes; sharp edges; excessive convexity
- KU22.** how to control distortion (such as welding sequence; deposition technique)
- KU23.** magnetic arc blow or arc deflection, causes and methods to avoid or compensate
- KU24.** Various testing techniques like visual, destructive and non-destructive
- KU25.** common welder testing codes i.e. ASME section IX, EN 287, ISO 9606, IS 7310 and their purpose
- KU26.** safety requirements during the welding work

## Qualification Pack

### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret drawings, work instructions, equipment manuals and process documents
- GS2.** communicate the welding process requirements to the supervisor and co-workers
- GS3.** attentively listen and comprehend the information given by the supervisor/team members
- GS4.** write any work related information in English/regional language
- GS5.** recognise a workplace problem and take suitable action
- GS6.** analyse and apply the information gathered from observation, experience, reasoning or communication to act efficiently
- GS7.** plan and organize tools, machines and consumables for carrying out welding job
- GS8.** complete the assigned tasks with minimum supervision
- GS9.** report to the supervisor or deal with a colleague individually, depending on the type of concern



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                              | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Prepare for welding operations</i>                                                                                                                                                                         | <b>13</b>    | <b>19</b>       | -             | <b>7</b>   |
| <b>PC1.</b> identify the work to be done and product specifications by interpreting the product drawing, Welding Procedure Specification (WPS) and job orders                                                 | 1            | 2               | -             | 1          |
| <b>PC2.</b> identify the tools, welding machines, measuring instruments, accessories, consumables and input materials (i.e. carbon, low alloy steel etc.) as per the requirements mentioned in WPS or drawing | 3            | 2               | -             | 2          |
| <b>PC3.</b> select and arrange the right material, equipment, fixtures, accessories, welding torch and consumables i.e. electrode, filler wire, shielding gas etc. as per the SOP and job requirements        | 2            | 3               | -             | 1          |
| <b>PC4.</b> check the input material, tools and equipment for any defects and that they are as per the required quality standards                                                                             | 2            | 4               | -             | 1          |
| <b>PC5.</b> prepare the work area for the welding activities                                                                                                                                                  | 1            | 1               | -             | -          |
| <b>PC6.</b> prepare the materials (i.e. plate(1.5 - 24mm)/ sheet (1.5mm)) and joint for welding process                                                                                                       | 1            | 1               | -             | 1          |
| <b>PC7.</b> set the MMAW machine and its parameters as per the WPS and SOP                                                                                                                                    | 1            | 2               | -             | 1          |
| <b>PC8.</b> re-dry electrodes as per electrode classification requirement                                                                                                                                     | 1            | 1               | -             | -          |
| <b>PC9.</b> install the work pieces and fixture on the apparatus and align them with the electrodes as per the job requirements                                                                               | 1            | 2               | -             | -          |
| <b>PC10.</b> verify set up by running test weld specimen (scrap plate)                                                                                                                                        | -            | 1               | -             | -          |
| <i>Perform MMAW/SMAW operations</i>                                                                                                                                                                           | <b>11</b>    | <b>20</b>       | -             | <b>8</b>   |
| <b>PC11.</b> follow safety precautions during welding work as per SOP and organizational guidelines                                                                                                           | -            | 1               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                                                   | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC12.</b> start the MMAW machine for welding operations                                                                                                                                                         | 1            | 2               | -             | -          |
| <b>PC13.</b> strike and maintain a stable arc by applying correct technique (i.e. scratch start, tapping techniques) and to avoid welding defects                                                                  | 1            | 2               | -             | 1          |
| <b>PC14.</b> perform MMAW welding process as per SOP and tack weld the joint at appropriate intervals to produce joints of the specified quality, dimensions and profile                                           | 2            | 4               | -             | 2          |
| <b>PC15.</b> produce fillet and grove joints in 1F/1G, 2F/2G and 3F/ 3G welding positions as specified in WPS by using single or multi-run welds                                                                   | 2            | 4               | -             | 2          |
| <b>PC16.</b> ensure correct angle of torch, travel speed, direction of weld and feed during the welding operation                                                                                                  | 1            | 1               | -             | 1          |
| <b>PC17.</b> maintain proper bead sequence with respect to groove/fillet configurations and positions                                                                                                              | -            | 1               | -             | -          |
| <b>PC18.</b> monitor the welding process parameters (air pressure, electrode force, electrode distance, gas flow etc.) are within standards by reading the various gauges and correct them if not within standards | 1            | 1               | -             | 1          |
| <b>PC19.</b> measure the final welded piece and compare with the dimensional and geometrical aspects of the weld as prescribed in the WPS and engineering drawing                                                  | 1            | 1               | -             | 1          |
| <b>PC20.</b> remove extra material, slag etc. by using brush, chipping hammers, grinders etc., from the welded piece                                                                                               | 1            | 1               | -             | -          |
| <b>PC21.</b> hammer the work piece to get the desired shape, if there are any welding bulges/distortions                                                                                                           | 1            | 1               | -             | -          |
| <b>PC22.</b> shut down the welding equipment and remove the workpiece after completion of welding activities                                                                                                       | -            | 1               | -             | -          |
| <i>Perform post-welding operations</i>                                                                                                                                                                             | <b>6</b>     | <b>11</b>       | -             | <b>5</b>   |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                     | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC23.</b> check the work pieces as per the work instructions for product quality                                                  | 1            | 2               | -             | 1          |
| <b>PC24.</b> identify various weld defects by conducting visual inspection, destructive and non-destructive tests on the work pieces | 2            | 4               | -             | 2          |
| <b>PC25.</b> separate the defective pieces which can be repaired/ reworked and pieces which are beyond repair                        | 1            | 1               | -             | 1          |
| <b>PC26.</b> clean and store all the tools, machine and equipment after completion of work                                           | 1            | 2               | -             | 1          |
| <b>PC27.</b> dispose scrap or waste material in accordance with the company policies and environmental regulations                   | 1            | 1               | -             | -          |
| <b>PC28.</b> report to the supervisor about any problems faced or anticipated during the complete process                            | -            | 1               | -             | -          |
| <b>NOS Total</b>                                                                                                                     | <b>30</b>    | <b>50</b>       | <b>-</b>      | <b>20</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                                                                                                                         |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NOS Code</b>            | CSC/N0204                                                                                                                                                               |
| <b>NOS Name</b>            | Manually weld carbon and low alloy steels by using Metal Arc Welding (MMAW)/ Shielded Metal Arc Welding (SMAW)                                                          |
| <b>Sector</b>              | Capital Goods                                                                                                                                                           |
| <b>Sub-Sector</b>          | Machine Tools, Dies, Moulds and Press Tools, Plastics Manufacturing Machinery, Textile Manufacturing Machinery, Process Plant Machinery, Electrical and Power Machinery |
| <b>Occupation</b>          | Welding and Cutting                                                                                                                                                     |
| <b>NSQF Level</b>          | 3                                                                                                                                                                       |
| <b>Credits</b>             | TBD                                                                                                                                                                     |
| <b>Version</b>             | 2.0                                                                                                                                                                     |
| <b>Last Reviewed Date</b>  | 08/05/2025                                                                                                                                                              |
| <b>Next Review Date</b>    | 08/05/2028                                                                                                                                                              |
| <b>NSQF Clearance Date</b> | 08/05/2025                                                                                                                                                              |

## Qualification Pack

### CSC/N0209: Manually weld metals by using MIG/MAG welding

#### Description

This unit is about performing MIG/MAG welding for producing various types of joints on metal and metal alloys as per the given specifications and standards specified by the organisation.

#### Scope

The scope covers the following :

- Prepare for welding operations
- Perform MIG/MAG welding operations
- Perform post-welding operations

#### Elements and Performance Criteria

##### *Prepare for welding operations*

To be competent, the user/individual on the job must be able to:

- PC1.** identify the work to be done and product specifications by interpreting the product drawing, Welding Procedure Specification (WPS) and job orders
- PC2.** identify the tools, MIG welding machines, measuring instruments, accessories, consumables and input materials (i.e. ferrous metals/materials: carbon steel, stainless steel etc.) as per the requirements mentioned in WPS or drawing
- PC3.** select and arrange the right material, equipment, fixtures, accessories, welding torch and consumables i.e. electrode, filler wire, shielding gas etc. as per the SOP and job requirements
- PC4.** check the input material, tools and equipment for any defects and that they are as per the required quality standards
- PC5.** prepare the work area for welding activities
- PC6.** prepare the materials (i.e. sheet (less than 1.5 mm), plate, structural section, pipe/tube, other forms) and joint for welding process
- PC7.** clean wire feeder and torch tip
- PC8.** set the MIG welding machine and its parameters i.e. wire feed rate, amperage, gas flow rate etc. as per the WPS and SOP
- PC9.** connect and adjust regulators and flow meters to cylinders
- PC10.** choose appropriate mode of metal transfer
- PC11.** set pre-purge with shielding gas as required
- PC12.** install the work pieces and fixture on the apparatus and align them with the electrodes as per the job requirements
- PC13.** verify set up by running test weld on the specimen (scrap plate)

##### *Perform MIG/MAG welding operations*

To be competent, the user/individual on the job must be able to:

- PC14.** follow safety precautions during welding work as per SOP and organizational guidelines
- PC15.** start the MIG welding machine for welding operations

## Qualification Pack

- PC16.** perform MIG welding process in all welding positions as per SOP and tack weld the joint at appropriate intervals to produce joints of the specified quality, dimensions and profile
- PC17.** adjust wire stick-out as per requirement
- PC18.** produce joints of the required quality and of specified dimensional accuracy which achieve a weld quality equivalent to Level C of ISO 5817
- PC19.** ensure correct angle of torch, travel speed, direction of weld and feed during the welding operation
- PC20.** monitor the welding process parameters (air pressure, electrode force, electrode distance, gas flow etc.) are within standards by reading the various gauges and correct them if not within standards
- PC21.** measure the final welded piece and compare with the dimensions as prescribed in the WPS and engineering drawing
- PC22.** remove extra material, slag etc. by using brush, chipping hammers, grinders etc., from the welded piece
- PC23.** shut down the welding equipment and remove the workpiece after completion of welding activities

### *Perform post-welding operations*

To be competent, the user/individual on the job must be able to:

- PC24.** check the work pieces as per the work instructions for product quality
- PC25.** identify various weld defects by conducting visual inspection, destructive and non-destructive tests on the work pieces
- PC26.** separate the defective pieces which can be repaired/ reworked and pieces which are beyond repair
- PC27.** clean and store all the tools, machine and equipment after completion of work
- PC28.** dispose scrap or waste material in accordance with the company policies and environmental regulations
- PC29.** check the machine operations for any malfunctions/defects in the component and immediately inform the supervisor/maintenance team for correction
- PC30.** report to the supervisor about any problems faced or anticipated during the complete process

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant legislation, standards, policies, and procedures followed in the organization
- KU2.** MIG welding and its process flow
- KU3.** various types of welding joints i.e. fillet lap joints, tee fillet joints, corner joints, butt joints (square, single, vee, double vee)
- KU4.** various welding positions i.e. flat (PA) 1G/1F, horizontal vertical (PB) 2F, horizontal (PC) 2G, vertical upwards (PF) 3F / 3G, vertical downwards (PG) 3F / 3G, Plate to Pipe (Fixed) 5F
- KU5.** how to read and interpret WPS, welding drawings and symbols

## Qualification Pack

- KU6.**
- welding specific equipment requirements for MIG/MAG welding
  - MIG equipment: rectifier (diode, thyristor/transistor), inverter, generator; wire feed system; measurement equipment for measuring; electrical output and continuity (voltmeter/multi-meter, ammeter/shunts/coils, tong tester); welding cables - wire feed to torch (air cooled, harness construction); welding guns/torches (air cooled, construction, types [push, pull, reel-on-gun] swan neck design, pistol design); nozzles (dip, spray); return clamps (types, clamping mechanisms) and cables; solenoid valves (shielding gas); jog-feed control, gas purge control; ancillary equipment (angle grinders, wire brushes, liners, hammer, power saw, angle, pedestal and straight grinders, chisel); other tools and equipment such as wrenches, wire cutters and MIG pliers
- KU7.** SOP recommended by the manufacturer for using tools, measuring instruments, accessories, MMAW welding machine etc. during the welding process
- KU8.** main components and controls of welding equipment
- KU9.** ISO colour codes for welding apparatus such as gas cylinder, hoses, electric cables, etc.
- KU10.** joint preparation process: made rust free; cleaned - free from scaling, paint, oil/grease; made dry and free from moisture; edges to be welded prepared as per job requirement - such as flat, square or bevelled; use various machines and techniques for the above (e.g. chamfering machine, grinding and stripping, gas or plasma cutting, etc.); correctly positioned (positioning: devices and techniques; jigs and fixtures; setting up joint in correct position & alignment)
- KU11.** impact of various welding parameters like voltage, current, gas flow rate, speed, pressure, torch angle, cycle time, electrode distance etc. on the quality and quantity of welding
- KU12.** relationship between wire feed, speed control and welding current
- KU13.** MIG/MAG welding technique: e.g. fine adjustment of parameters, correct manipulation of the torch, blending in stops/starts, tack welds, angle of the torch, setting of individual parameters like wire feed speed, voltage, gas flow rate, stick-out, etc. various materials used for GMAW welding and their properties
- KU14.** SOP recommended by the organisation for operating MIG welding machine and its accessories
- KU15.** current and polarity required for GMAW
- KU16.** types, selection and application of filler wires and welding electrodes
- KU17.**
- reasons for using shielding gases, and the types and application of the various gases
  - Shielding gases: applications for shielding gases/gas mixtures (argon, mixture, helium, argon/helium mixtures, helium/argon mixtures, argon/hydrogen mixtures, nitrogen argon/nitrogen mixtures, CO<sub>2</sub> and CO<sub>2</sub> mixtures); flow rates for applications; identify percentage of purity and mixture with respect to WPS/PQR
- KU18.**
- use, impact and importance of gas pressures and flow rates (in relationship to the type of material being welded)
  - Types of ferrous metals/materials: carbon steel, stainless steel
- KU19.**
- methods/modes of metal transfer and their uses
  - Methods: globular, short circuit transfer, spray arc, pulse, surface tension transfer (STT)
- KU20.** purpose and correct use of anti-spatter compound
- KU21.** importance and procedure to clean torch tip and liner
- KU22.**
- factors that determine weld bead shape
  - Factors: gun angles and weld bead profiles (push, perpendicular, drag); electrode extensions stick out (short, normal, long); fillet weld electrode extension stick out (short, normal, long); gun travel speed (slow, normal, fast); current and voltage

## Qualification Pack

- KU23.** • types of beads, characteristics and uses (stringer, weave, weave patterns)  
 • Bead characteristics: spatter deposits, roughness, evenness, fill, crater, overlap
- KU24.** • weld bead quality characteristics  
 • Bead characteristics: spatter deposits, roughness, evenness, fill, crater, overlap, contour – convex, concave, mitre
- KU25.** SOP recommended by the organisation for checking irregularities in the product/work piece
- KU26.** • factors that affect weld quality standards  
 • Quality standards: required parameters for dimensional accuracy; weld finishes are built up to the full section of the weld; joins at stop/start positions merge smoothly; weld surface is (free from cracks; substantially free from porosity; free from any pronounced hump or crater; substantially free from shrinkage cavities; substantially free from trapped slag; substantially free from arcing or chipping marks); fillet welds are (equal in leg length, slightly convex in profile (where applicable), size of the fillet equivalent to the thickness of the material welded); weld contour is (of linear and of uniform profile; smooth and free from excessive undulations; regular and has an even ripple formations); welds are adequately fused, and there is minimal undercut, overlap and surface inclusions etc.
- KU27.** various defects associated with the MIG welding process
- KU28.** how to control distortion (such as welding sequence; deposition technique)
- KU29.** various testing techniques like visual, destructive and non-destructive
- KU30.** safety requirements during the welding work

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret drawings, work instructions, equipment manuals and process documents
- GS2.** communicate the welding process requirements to the supervisor and co-workers
- GS3.** attentively listen and comprehend the information given by the supervisor/team members
- GS4.** write any work related information in English/regional language
- GS5.** recognise a workplace problem and take suitable action
- GS6.** analyse and apply the information gathered from observation, experience, reasoning or communication to act efficiently
- GS7.** plan and organize tools, machines and consumables for carrying out welding job
- GS8.** complete the assigned tasks with minimum supervision
- GS9.** report to the supervisor or deal with a colleague individually, depending on the type of concern



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                  | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Prepare for welding operations</i>                                                                                                                                                                                                             | <b>15</b>    | <b>21</b>       | -             | <b>7</b>   |
| <b>PC1.</b> identify the work to be done and product specifications by interpreting the product drawing, Welding Procedure Specification (WPS) and job orders                                                                                     | 1            | 2               | -             | 1          |
| <b>PC2.</b> identify the tools, MIG welding machines, measuring instruments, accessories, consumables and input materials (i.e. ferrous metals/materials: carbon steel, stainless steel etc.) as per the requirements mentioned in WPS or drawing | 3            | 2               | -             | 1          |
| <b>PC3.</b> select and arrange the right material, equipment, fixtures, accessories, welding torch and consumables i.e. electrode, filler wire, shielding gas etc. as per the SOP and job requirements                                            | 2            | 3               | -             | 1          |
| <b>PC4.</b> check the input material, tools and equipment for any defects and that they are as per the required quality standards                                                                                                                 | 2            | 3               | -             | 1          |
| <b>PC5.</b> prepare the work area for welding activities                                                                                                                                                                                          | 1            | 1               | -             | -          |
| <b>PC6.</b> prepare the materials (i.e. sheet (less than 1.5 mm), plate, structural section, pipe/tube, other forms) and joint for welding process                                                                                                | 1            | 1               | -             | 1          |
| <b>PC7.</b> clean wire feeder and torch tip                                                                                                                                                                                                       | -            | 1               | -             | -          |
| <b>PC8.</b> set the MIG welding machine and its parameters i.e. wire feed rate, amperage, gas flow rate etc. as per the WPS and SOP                                                                                                               | 1            | 2               | -             | 1          |
| <b>PC9.</b> connect and adjust regulators and flow meters to cylinders                                                                                                                                                                            | 1            | 1               | -             | -          |
| <b>PC10.</b> choose appropriate mode of metal transfer                                                                                                                                                                                            | 1            | 1               | -             | -          |
| <b>PC11.</b> set pre-purge with shielding gas as required                                                                                                                                                                                         | 1            | 1               | -             | 1          |
| <b>PC12.</b> install the work pieces and fixture on the apparatus and align them with the electrodes as per the job requirements                                                                                                                  | 1            | 2               | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                                                   | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC13.</b> verify set up by running test weld on the specimen (scrap plate)                                                                                                                                      | -            | 1               | -             | -          |
| <i>Perform MIG/MAG welding operations</i>                                                                                                                                                                          | <b>8</b>     | <b>17</b>       | -             | <b>8</b>   |
| <b>PC14.</b> follow safety precautions during welding work as per SOP and organizational guidelines                                                                                                                | -            | 1               | -             | -          |
| <b>PC15.</b> start the MIG welding machine for welding operations                                                                                                                                                  | 1            | 2               | -             | -          |
| <b>PC16.</b> perform MIG welding process in all welding positions as per SOP and tack weld the joint at appropriate intervals to produce joints of the specified quality, dimensions and profile                   | 2            | 4               | -             | 2          |
| <b>PC17.</b> adjust wire stick-out as per requirement                                                                                                                                                              | 1            | 1               | -             | 1          |
| <b>PC18.</b> produce joints of the required quality and of specified dimensional accuracy which achieve a weld quality equivalent to Level C of ISO 5817                                                           | 1            | 4               | -             | 2          |
| <b>PC19.</b> ensure correct angle of torch, travel speed, direction of weld and feed during the welding operation                                                                                                  | 1            | 1               | -             | 1          |
| <b>PC20.</b> monitor the welding process parameters (air pressure, electrode force, electrode distance, gas flow etc.) are within standards by reading the various gauges and correct them if not within standards | 1            | 1               | -             | 1          |
| <b>PC21.</b> measure the final welded piece and compare with the dimensions as prescribed in the WPS and engineering drawing                                                                                       | 1            | 1               | -             | 1          |
| <b>PC22.</b> remove extra material, slag etc. by using brush, chipping hammers, grinders etc., from the welded piece                                                                                               | -            | 1               | -             | -          |
| <b>PC23.</b> shut down the welding equipment and remove the workpiece after completion of welding activities                                                                                                       | -            | 1               | -             | -          |
| <i>Perform post-welding operations</i>                                                                                                                                                                             | <b>7</b>     | <b>12</b>       | -             | <b>5</b>   |
| <b>PC24.</b> check the work pieces as per the work instructions for product quality                                                                                                                                | 1            | 2               | -             | 1          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                              | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC25.</b> identify various weld defects by conducting visual inspection, destructive and non-destructive tests on the work pieces                          | 2            | 3               | -             | 2          |
| <b>PC26.</b> separate the defective pieces which can be repaired/ reworked and pieces which are beyond repair                                                 | 1            | 1               | -             | 1          |
| <b>PC27.</b> clean and store all the tools, machine and equipment after completion of work                                                                    | 1            | 2               | -             | 1          |
| <b>PC28.</b> dispose scrap or waste material in accordance with the company policies and environmental regulations                                            | 1            | 1               | -             | -          |
| <b>PC29.</b> check the machine operations for any malfunctions/defects in the component and immediately inform the supervisor/maintenance team for correction | 1            | 2               | -             | -          |
| <b>PC30.</b> report to the supervisor about any problems faced or anticipated during the complete process                                                     | -            | 1               | -             | -          |
| <b>NOS Total</b>                                                                                                                                              | <b>30</b>    | <b>50</b>       | <b>-</b>      | <b>20</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                                                                                                                                                  |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NOS Code</b>            | CSC/N0209                                                                                                                                                                                        |
| <b>NOS Name</b>            | Manually weld metals by using MIG/MAG welding                                                                                                                                                    |
| <b>Sector</b>              | Capital Goods                                                                                                                                                                                    |
| <b>Sub-Sector</b>          | Machine Tools, Dies, Moulds and Press Tools, Plastics Manufacturing Machinery, Textile Manufacturing Machinery, Process Plant Machinery, Electrical and Power Machinery, Light Engineering Goods |
| <b>Occupation</b>          | Welding and Cutting                                                                                                                                                                              |
| <b>NSQF Level</b>          | 4                                                                                                                                                                                                |
| <b>Credits</b>             | TBD                                                                                                                                                                                              |
| <b>Version</b>             | 2.0                                                                                                                                                                                              |
| <b>Last Reviewed Date</b>  | 08/05/2025                                                                                                                                                                                       |
| <b>Next Review Date</b>    | 08/05/2028                                                                                                                                                                                       |
| <b>NSQC Clearance Date</b> | 08/05/2025                                                                                                                                                                                       |

## Qualification Pack

### DGT/VSQ/N0101: Employability Skills (30 Hours)

#### Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

#### Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

#### Elements and Performance Criteria

##### *Introduction to Employability Skills*

To be competent, the user/individual on the job must be able to:

**PC1.** understand the significance of employability skills in meeting the job requirements

##### *Constitutional values – Citizenship*

To be competent, the user/individual on the job must be able to:

**PC2.** identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

##### *Becoming a Professional in the 21st Century*

To be competent, the user/individual on the job must be able to:

**PC3.** explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

##### *Basic English Skills*

To be competent, the user/individual on the job must be able to:

**PC4.** speak with others using some basic English phrases or sentences

##### *Communication Skills*

To be competent, the user/individual on the job must be able to:

**PC5.** follow good manners while communicating with others

**PC6.** work with others in a team

## Qualification Pack

### *Diversity & Inclusion*

To be competent, the user/individual on the job must be able to:

**PC7.** communicate and behave appropriately with all genders and PwD

**PC8.** report any issues related to sexual harassment

### *Financial and Legal Literacy*

To be competent, the user/individual on the job must be able to:

**PC9.** use various financial products and services safely and securely

**PC10.** calculate income, expenses, savings etc.

**PC11.** approach the concerned authorities for any exploitation as per legal rights and laws

### *Essential Digital Skills*

To be competent, the user/individual on the job must be able to:

**PC12.** operate digital devices and use its features and applications securely and safely

**PC13.** use internet and social media platforms securely and safely

### *Entrepreneurship*

To be competent, the user/individual on the job must be able to:

**PC14.** identify and assess opportunities for potential business

**PC15.** identify sources for arranging money and associated financial and legal challenges

### *Customer Service*

To be competent, the user/individual on the job must be able to:

**PC16.** identify different types of customers

**PC17.** identify customer needs and address them appropriately

**PC18.** follow appropriate hygiene and grooming standards

### *Getting ready for apprenticeship & Jobs*

To be competent, the user/individual on the job must be able to:

**PC19.** create a basic biodata

**PC20.** search for suitable jobs and apply

**PC21.** identify and register apprenticeship opportunities as per requirement

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** need for employability skills

**KU2.** various constitutional and personal values

**KU3.** different environmentally sustainable practices and their importance

**KU4.** Twenty first (21st) century skills and their importance

**KU5.** how to use basic spoken English language

**KU6.** Do and dont of effective communication

**KU7.** inclusivity and its importance

**KU8.** different types of disabilities and appropriate communication and behaviour towards PwD

**KU9.** different types of financial products and services

## Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                               | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Introduction to Employability Skills</i>                                                                                                                                                                                                                    | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC1.</b> understand the significance of employability skills in meeting the job requirements                                                                                                                                                                | -            | -               | -             | -          |
| <i>Constitutional values – Citizenship</i>                                                                                                                                                                                                                     | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC2.</b> identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices                                                                                                                         | -            | -               | -             | -          |
| <i>Becoming a Professional in the 21st Century</i>                                                                                                                                                                                                             | <b>1</b>     | <b>3</b>        | -             | -          |
| <b>PC3.</b> explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc. | -            | -               | -             | -          |
| <i>Basic English Skills</i>                                                                                                                                                                                                                                    | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC4.</b> speak with others using some basic English phrases or sentences                                                                                                                                                                                    | -            | -               | -             | -          |
| <i>Communication Skills</i>                                                                                                                                                                                                                                    | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC5.</b> follow good manners while communicating with others                                                                                                                                                                                                | -            | -               | -             | -          |
| <b>PC6.</b> work with others in a team                                                                                                                                                                                                                         | -            | -               | -             | -          |
| <i>Diversity &amp; Inclusion</i>                                                                                                                                                                                                                               | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC7.</b> communicate and behave appropriately with all genders and PwD                                                                                                                                                                                      | -            | -               | -             | -          |
| <b>PC8.</b> report any issues related to sexual harassment                                                                                                                                                                                                     | -            | -               | -             | -          |
| <i>Financial and Legal Literacy</i>                                                                                                                                                                                                                            | <b>3</b>     | <b>4</b>        | -             | -          |
| <b>PC9.</b> use various financial products and services safely and securely                                                                                                                                                                                    | -            | -               | -             | -          |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                  | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC10.</b> calculate income, expenses, savings etc.                                             | -            | -               | -             | -          |
| <b>PC11.</b> approach the concerned authorities for any exploitation as per legal rights and laws | -            | -               | -             | -          |
| <i>Essential Digital Skills</i>                                                                   | <b>4</b>     | <b>6</b>        | -             | -          |
| <b>PC12.</b> operate digital devices and use its features and applications securely and safely    | -            | -               | -             | -          |
| <b>PC13.</b> use internet and social media platforms securely and safely                          | -            | -               | -             | -          |
| <i>Entrepreneurship</i>                                                                           | <b>3</b>     | <b>5</b>        | -             | -          |
| <b>PC14.</b> identify and assess opportunities for potential business                             | -            | -               | -             | -          |
| <b>PC15.</b> identify sources for arranging money and associated financial and legal challenges   | -            | -               | -             | -          |
| <i>Customer Service</i>                                                                           | <b>2</b>     | <b>2</b>        | -             | -          |
| <b>PC16.</b> identify different types of customers                                                | -            | -               | -             | -          |
| <b>PC17.</b> identify customer needs and address them appropriately                               | -            | -               | -             | -          |
| <b>PC18.</b> follow appropriate hygiene and grooming standards                                    | -            | -               | -             | -          |
| <i>Getting ready for apprenticeship &amp; Jobs</i>                                                | <b>1</b>     | <b>3</b>        | -             | -          |
| <b>PC19.</b> create a basic biodata                                                               | -            | -               | -             | -          |
| <b>PC20.</b> search for suitable jobs and apply                                                   | -            | -               | -             | -          |
| <b>PC21.</b> identify and register apprenticeship opportunities as per requirement                | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                  | <b>20</b>    | <b>30</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                 |
|----------------------------|---------------------------------|
| <b>NOS Code</b>            | DGT/VSQ/N0101                   |
| <b>NOS Name</b>            | Employability Skills (30 Hours) |
| <b>Sector</b>              | Cross Sectoral                  |
| <b>Sub-Sector</b>          | Professional Skills             |
| <b>Occupation</b>          | Employability                   |
| <b>NSQF Level</b>          | 2                               |
| <b>Credits</b>             | 1                               |
| <b>Version</b>             | 1.0                             |
| <b>Last Reviewed Date</b>  | 12/03/2026                      |
| <b>Next Review Date</b>    | 12/09/2026                      |
| <b>NSQC Clearance Date</b> | 12/03/2026                      |

## Qualification Pack

### CSC/N0215: Perform AR-guided SMAW/GMAW in Oil & Gas Industry

#### Description

This toolkit should leverage AR to overlay instructions, measurements and visual cues onto the real world welding environments, improving accuracy & safety.

#### Scope

The scope covers the following :

- Prepare for AR-Guided Oil & Gas Setup
- Perform AR Based Oil & Gas Welding operations
- Perform post-AR Guided welding operations

#### Elements and Performance Criteria

##### *Prepare for AR-Guided Oil & Gas Setup*

To be competent, the user/individual on the job must be able to:

- PC1.** Get awareness regarding working of simulator which integrates 3D models of pipes, oil & gas pipelines
- PC2.** Ensure accurate measurements and angles during welding preparation and execution using AR tools.
- PC3.** Install and open AR software that provides welding instructions, visual aids, and measurement tools specific to the Oil & Gas Industrial project.
- PC4.** Perform calibration on AR devices to ensure accurate alignment of digital overlays with physical components.
- PC5.** Perform AR overlays that show the weld beep and formation of heat etc.

##### *Perform AR Based Oil & Gas Welding operations*

To be competent, the user/individual on the job must be able to:

- PC6.** Charge and test AR devices (e.g., smart glasses, tablets) and ensure that software applications are updated and functioning properly.
- PC7.** Access project-specific data, such as welding specifications, diagrams, and AR-guided instructions relevant to the task.
- PC8.** Follow AR-guided instructions to properly clean, fit, and align the surfaces to be welded.
- PC9.** Utilize AR overlays that provide step-by-step guidance on welding techniques, acceptable angles, and movement patterns.
- PC10.** Ensure accurate measurements during welding preparations and executions using AR tools.

##### *Perform post-AR Guided welding operations*

To be competent, the user/individual on the job must be able to:

- PC11.** Allow welded joints to cool naturally or use appropriate cooling methods following the welding process to prevent warping or cracking.
- PC12.** Perform a preliminary visual inspection of the welds for burn-through, undercuts, or other visible defects.

## Qualification Pack

**PC13.** Follow step by step instructions through AR overlays and get real time feedback to welding parameters/techniques.

### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Technical Knowledge of Welding Processes
- KU2.** Types of oil & Gas welding (e.g., SMAW, TIG, MIG, FCAW)
- KU3.** Welding materials and consumables
- KU4.** Heat control and metallurgy in welding
- KU5.** Pre-weld and post-weld procedures
- KU6.** Pipe Welding Specifics
- KU7.** Basic rail track components (rails, sleepers, joints, fasteners)
- KU8.** Understanding of Pipe alignment and leveling
- KU9.** Knowledge Materials & Metallurgy
- KU10.** Safety Protocols and Hazard Awareness
- KU11.** Personal Protective Equipment (PPE) requirements
- KU12.** Fire and chemical handling safety (especially with Thermit welding)
- KU13.** Emergency procedures in a welding environment
- KU14.** AR System Operation and Interpretation
- KU15.** How to use AR devices (HoloLens, AR headsets, or tablets)
- KU16.** Understanding of AR overlays, prompts, and virtual guides
- KU17.** Ability to follow step-by-step digital instructions
- KU18.** Tool and Equipment Familiarity
- KU19.** Identification and correct use of welding tools (grinders, clamps, crucibles)
- KU20.** Maintenance and inspection of tools before/after use
- KU21.** Virtual simulation of tool use via AR
- KU22.** AR Software Proficiency: Skills in navigating AR applications for accessing and manipulating welding instructions and project data

### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret drawings, work instructions, equipment manuals and process documents
- GS2.** communicate the welding process requirements to the supervisor and co-workers
- GS3.** attentively listen and comprehend the information given by the supervisor/team members
- GS4.** write any work related information in English/regional language
- GS5.** recognize a workplace problem and take suitable action
- GS6.** analyze and apply the information gathered from observation, experience, reasoning or communication to act efficiently

## Qualification Pack

- GS7.** plan and organize tools, machines and consumables for carrying out welding job
- GS8.** complete the assigned tasks with minimum supervision
- GS9.** report to the supervisor or deal with a colleague individually, depending on the type of concern

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                              | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Prepare for AR-Guided Oil &amp; Gas Setup</i>                                                                                                              | <b>10</b>    | <b>15</b>       | -             | <b>7</b>   |
| <b>PC1.</b> Get awareness regarding working of simulator which integrates 3D models of pipes, oil & gas pipelines                                             | 1            | 1               | -             | -          |
| <b>PC2.</b> Ensure accurate measurements and angles during welding preparation and execution using AR tools.                                                  | 2            | 4               | -             | 2          |
| <b>PC3.</b> Install and open AR software that provides welding instructions, visual aids, and measurement tools specific to the Oil & Gas Industrial project. | 2            | 4               | -             | 2          |
| <b>PC4.</b> Perform calibration on AR devices to ensure accurate alignment of digital overlays with physical components.                                      | 3            | 4               | -             | 2          |
| <b>PC5.</b> Perform AR overlays that show the weld beep and formation of heat etc.                                                                            | 2            | 2               | -             | 1          |
| <i>Perform AR Based Oil &amp; Gas Welding operations</i>                                                                                                      | <b>10</b>    | <b>15</b>       | -             | <b>8</b>   |
| <b>PC6.</b> Charge and test AR devices (e.g., smart glasses, tablets) and ensure that software applications are updated and functioning properly.             | 2            | 2               | -             | 3          |
| <b>PC7.</b> Access project-specific data, such as welding specifications, diagrams, and AR-guided instructions relevant to the task.                          | 2            | 4               | -             | 2          |
| <b>PC8.</b> Follow AR-guided instructions to properly clean, fit, and align the surfaces to be welded.                                                        | 2            | 4               | -             | 1          |
| <b>PC9.</b> Utilize AR overlays that provide step-by-step guidance on welding techniques, acceptable angles, and movement patterns.                           | 2            | 3               | -             | 2          |
| <b>PC10.</b> Ensure accurate measurements during welding preparations and executions using AR tools.                                                          | 2            | 2               | -             | -          |
| <i>Perform post-AR Guided welding operations</i>                                                                                                              | <b>10</b>    | <b>20</b>       | -             | <b>5</b>   |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                    | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC11.</b> Allow welded joints to cool naturally or use appropriate cooling methods following the welding process to prevent warping or cracking. | 4            | 5               | -             | 1          |
| <b>PC12.</b> Perform a preliminary visual inspection of the welds for burn-through, undercuts, or other visible defects.                            | 3            | 8               | -             | 3          |
| <b>PC13.</b> Follow step by step instructions through AR overlays and get real time feedback to welding parameters/techniques.                      | 3            | 7               | -             | 1          |
| <b>NOS Total</b>                                                                                                                                    | <b>30</b>    | <b>50</b>       | <b>-</b>      | <b>20</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                   |
|----------------------------|---------------------------------------------------|
| <b>NOS Code</b>            | CSC/N0215                                         |
| <b>NOS Name</b>            | Perform AR-guided SMAW/GMAW in Oil & Gas Industry |
| <b>Sector</b>              | Capital Goods                                     |
| <b>Sub-Sector</b>          |                                                   |
| <b>Occupation</b>          | Welding and Cutting                               |
| <b>NSQF Level</b>          | 4                                                 |
| <b>Credits</b>             | 1                                                 |
| <b>Version</b>             | 1.0                                               |
| <b>Last Reviewed Date</b>  | 08/05/2025                                        |
| <b>Next Review Date</b>    | 08/05/2028                                        |
| <b>NSQC Clearance Date</b> | 08/05/2025                                        |



## Qualification Pack

### CSC/N0216: Perform AR-guided SMAW/GMAW in Railways

#### Description

An AR welding simulator provides a virtual, interactive training experience that enhances learning & skill development in welding. It overlays digital welding techniques onto real world environment allowing learners to practice and refine their skills in a safe controlled setting.

#### Scope

The scope covers the following :

- Prepare for AR-Guided Oil & Gas Setup
- Perform AR Based Oil & Gas Welding operations
- Perform post-AR Guided welding operations

#### Elements and Performance Criteria

##### *Weld Preparation & Set Up*

To be competent, the user/individual on the job must be able to:

- PC1.** Utilize the AR enabled Tablet/Headset overlays for instruction, measurements & visual cues for joint preparation (eg; bevel angles, root gap etc.).
- PC2.** Perform real time measurements of weld joint dimensions using AR tools.
- PC3.** Follow AR guidelines for setting up of welding equipment.
- PC4.** Utilize AR tools to identify & mark areas required for cleaning & grinding
- PC5.** Follow AR guidelines for application of appropriate flux cleansing agents
- PC6.** Utilize AR overlays to ensure correct alignment of weld joints and components

##### *Welding Execution*

To be competent, the user/individual on the job must be able to:

- PC7.** Utilize AR overlays for welding parameters (eg: voltage, amperage, wire speed) and settings onto welding environment.
- PC8.** Utilize AR overlays that provide step-by-step guidance on welding techniques, acceptable angles, and movement patterns.
- PC9.** Ensure accurate measurements during welding preparations and executions using AR tools.

##### *Perform post-AR Guided welding operations*

To be competent, the user/individual on the job must be able to:

- PC10.** Allow welded joints to cool naturally or use appropriate cooling methods following the welding process to prevent warping or cracking.
- PC11.** Utilize AR inspection tools overlay for visual inspection of welds (inspection points & reference lines).
- PC12.** Utilize AR overlay measurement tools to accurately measure weld dimensions (eg: weld height, width).
- PC13.** Follow step by step instructions through AR overlays and get real time feedback to welding parameters/techniques.

## Qualification Pack

### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Technical Knowledge of Welding Processes
- KU2.** Types of Railways welding (e.g., SMAW, TIG, MIG, FCAW)
- KU3.** Welding materials and consumables
- KU4.** Heat control and metallurgy in welding
- KU5.** Pre-weld and post-weld procedures
- KU6.** Purpose of visual cues and benefits of AR guided welding
- KU7.** Basic rail track components (rails, sleepers, joints, fasteners)
- KU8.** Understanding of Pipe alignment and leveling
- KU9.** Knowledge Materials & Metallurgy
- KU10.** Safety Protocols and Hazard Awareness
- KU11.** Personal Protective Equipment (PPE) requirements
- KU12.** Fire and chemical handling safety (especially with Thermit welding)
- KU13.** Emergency procedures in a welding environment
- KU14.** AR System Operation and Interpretation
- KU15.** How to use AR devices (HoloLens, AR headsets, or tablets)
- KU16.** Understanding of AR overlays, prompts, and virtual guides
- KU17.** Ability to follow step-by-step digital instructions
- KU18.** Tool and Equipment Familiarity
- KU19.** Identification and correct use of welding tools (grinders, clamps, crucibles)
- KU20.** Maintenance and inspection of tools before/after use
- KU21.** Virtual simulation of tool use via AR
- KU22.** AR Software Proficiency: Skills in navigating AR applications for accessing and manipulating welding instructions and project data.

### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret drawings, work instructions, equipment manuals and process documents
- GS2.** communicate the welding process requirements to the supervisor and co-workers
- GS3.** attentively listen and comprehend the information given by the supervisor/team members
- GS4.** write any work related information in English/regional language
- GS5.** recognize a workplace problem and take suitable action
- GS6.** analyze and apply the information gathered from observation, experience, reasoning or communication to act efficiently
- GS7.** plan and organize tools, machines and consumables for carrying out welding job
- GS8.** complete the assigned tasks with minimum supervision

## Qualification Pack

- GS9.** report to the supervisor or deal with a colleague individually, depending on the type of concern

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Weld Preparation &amp; Set Up</i>                                                                                                                            | <b>10</b>    | <b>15</b>       | -             | <b>7</b>   |
| <b>PC1.</b> Utilize the AR enabled Tablet/Headset overlays for instruction, measurements & visual cues for joint preparation (eg; bevel angles, root gap etc.). | 1            | 1               | -             | -          |
| <b>PC2.</b> Perform real time measurements of weld joint dimensions using AR tools.                                                                             | 2            | 4               | -             | 2          |
| <b>PC3.</b> Follow AR guidelines for setting up of welding equipment.                                                                                           | 2            | 4               | -             | 2          |
| <b>PC4.</b> Utilize AR tools to identify & mark areas required for cleaning & grinding                                                                          | 3            | 4               | -             | 1          |
| <b>PC5.</b> Follow AR guidelines for application of appropriate flux cleansing agents                                                                           | 1            | 1               | -             | 1          |
| <b>PC6.</b> Utilize AR overlays to ensure correct alignment of weld joints and components                                                                       | 1            | 1               | -             | 1          |
| <i>Welding Execution</i>                                                                                                                                        | <b>10</b>    | <b>15</b>       | -             | <b>8</b>   |
| <b>PC7.</b> Utilize AR overlays for welding parameters (eg: voltage, amperage, wire speed) and settings onto welding environment.                               | 4            | 5               | -             | 3          |
| <b>PC8.</b> Utilize AR overlays that provide step-by-step guidance on welding techniques, acceptable angles, and movement patterns.                             | 4            | 6               | -             | 3          |
| <b>PC9.</b> Ensure accurate measurements during welding preparations and executions using AR tools.                                                             | 2            | 4               | -             | 2          |
| <i>Perform post-AR Guided welding operations</i>                                                                                                                | <b>10</b>    | <b>20</b>       | -             | <b>5</b>   |
| <b>PC10.</b> Allow welded joints to cool naturally or use appropriate cooling methods following the welding process to prevent warping or cracking.             | 2            | 2               | -             | -          |
| <b>PC11.</b> Utilize AR inspection tools overlay for visual inspection of welds (inspection points & reference lines).                                          | 2            | 3               | -             | 2          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                               | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC12.</b> Utilize AR overlay measurement tools to accurately measure weld dimensions (eg: weld height, width).              | 3            | 8               | -             | 2          |
| <b>PC13.</b> Follow step by step instructions through AR overlays and get real time feedback to welding parameters/techniques. | 3            | 7               | -             | 1          |
| <b>NOS Total</b>                                                                                                               | <b>30</b>    | <b>50</b>       | <b>-</b>      | <b>20</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                         |
|----------------------------|-----------------------------------------|
| <b>NOS Code</b>            | CSC/N0216                               |
| <b>NOS Name</b>            | Perform AR-guided SMAW/GMAW in Railways |
| <b>Sector</b>              | Capital Goods                           |
| <b>Sub-Sector</b>          |                                         |
| <b>Occupation</b>          | Welding and Cutting                     |
| <b>NSQF Level</b>          | 4                                       |
| <b>Credits</b>             | 1                                       |
| <b>Version</b>             | 1.0                                     |
| <b>Last Reviewed Date</b>  | 08/05/2025                              |
| <b>Next Review Date</b>    | 08/05/2028                              |
| <b>NSQC Clearance Date</b> | 08/05/2025                              |

## Qualification Pack

### CSC/N0217: Perform AR-guided SMAW/GMAW in Shipbuilding

#### Description

An AR welding simulator provides a virtual, interactive training experience that enhances learning & skill development in welding. It overlays digital welding techniques onto real world environment allowing learners to practice and refine their skills in a safe controlled setting. An AR based toolkit for SMAW/GMAW welding in shipbuilding should include tools and accessories for both welding processes, including power sources, welding guns/electrodes, safety equipment and inspection finishing tools all integrated with an AR application.

#### Scope

The scope covers the following :

- Prepare for AR-Guided Ship Building Setup
- Perform AR Based Ship Building Welding operations
- Perform post-AR Guided welding operations

#### Elements and Performance Criteria

##### *Weld Preparation & Set Up*

To be competent, the user/individual on the job must be able to:

- PC1.** Utilize the AR enabled Tablet/Headset overlays for instruction, measurements & visual cues for joint preparation.
- PC2.** Perform real time measurements of weld joint dimensions using AR tools.
- PC3.** Follow AR guidelines for setting up of welding equipment.
- PC4.** Follow AR guidelines for application of appropriate flux cleansing agents
- PC5.** Utilize AR equipment for performing welds in different positions (AR application should be adaptable to various welding positions & offer guidance accordingly).

##### *Perform AR Based Shipbuilding Welding Operations*

To be competent, the user/individual on the job must be able to:

- PC6.** Utilize AR overlays for displaying ideal weld parameters, position and techniques.
- PC7.** Utilize AR overlays that provide step-by-step guidance on welding techniques, acceptable angles, and movement patterns.
- PC8.** Ensure accurate measurements during welding preparations and executions using AR tools on AR Based projections of weld joints, shapes & sizes for accurate placement.

##### *Perform post-AR Guided welding operations*

To be competent, the user/individual on the job must be able to:

- PC9.** Allow welded joints to cool naturally or use appropriate cooling methods following the welding process to prevent warping or cracking.
- PC10.** Utilize AR inspection tools overlay for visual inspection of welds.
- PC11.** Utilize AR sensors & cameras for providing feedback on weld quality, heat input & other critical parameters and other feedback.

## Qualification Pack

### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Technical Knowledge of Welding Processes
- KU2.** Structural Components - Plates, hulls , bulkheads, frames
- KU3.** Alignment & Fit-Up - Importance of precision in joining large steel parts in marine structures.
- KU4.** Thermal Distortion & Control -Understanding heat effects on large metal surfaces and compensating techniques.
- KU5.** Pre-weld and post-weld procedures
- KU6.** Pipe Welding Specifics
- KU7.** Basic Shipbuilding components
- KU8.** Understanding of Pipe alignment and leveling
- KU9.** Knowledge Materials & Metallurgy
- KU10.** Safety Protocols and Hazard Awareness
- KU11.** Personal Protective Equipment (PPE) requirements
- KU12.** Fire and chemical handling safety (especially with Thermit welding)
- KU13.** Emergency procedures in a welding environment
- KU14.** AR System Operation and Interpretation
- KU15.** How to use AR devices (HoloLens, AR headsets, or tablets)
- KU16.** Understanding of AR overlays, prompts, and virtual guides
- KU17.** Ability to follow step-by-step digital instructions
- KU18.** Tool and Equipment Familiarity
- KU19.** Identification and correct use of welding tools (grinders, clamps, crucibles)
- KU20.** Maintenance and inspection of tools before/after use
- KU21.** Virtual simulation of tool use via AR
- KU22.** AR Software Proficiency: Skills in navigating AR applications for accessing and manipulating welding instructions and project data.

### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret drawings, work instructions, equipment manuals and process documents
- GS2.** communicate the welding process requirements to the supervisor and co-workers
- GS3.** attentively listen and comprehend the information given by the supervisor/team members
- GS4.** write any work related information in English/regional language
- GS5.** recognize a workplace problem and take suitable action
- GS6.** analyze and apply the information gathered from observation, experience, reasoning or communication to act efficiently
- GS7.** plan and organize tools, machines and consumables for carrying out welding job
- GS8.** complete the assigned tasks with minimum supervision



## Qualification Pack

- GS9.** report to the supervisor or deal with a colleague individually, depending on the type of concern

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                  | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Weld Preparation &amp; Set Up</i>                                                                                                                                              | <b>10</b>    | <b>15</b>       | -             | <b>7</b>   |
| <b>PC1.</b> Utilize the AR enabled Tablet/Headset overlays for instruction, measurements & visual cues for joint preparation.                                                     | 1            | 1               | -             | 1          |
| <b>PC2.</b> Perform real time measurements of weld joint dimensions using AR tools.                                                                                               | 2            | 4               | -             | 2          |
| <b>PC3.</b> Follow AR guidelines for setting up of welding equipment.                                                                                                             | 2            | 4               | -             | 2          |
| <b>PC4.</b> Follow AR guidelines for application of appropriate flux cleansing agents                                                                                             | 3            | 4               | -             | 1          |
| <b>PC5.</b> Utilize AR equipment for performing welds in different positions (AR application should be adaptable to various welding positions & offer guidance accordingly).      | 2            | 2               | -             | 1          |
| <i>Perform AR Based Shipbuilding Welding Operations</i>                                                                                                                           | <b>10</b>    | <b>15</b>       | -             | <b>8</b>   |
| <b>PC6.</b> Utilize AR overlays for displaying ideal weld parameters, position and techniques.                                                                                    | 4            | 5               | -             | 3          |
| <b>PC7.</b> Utilize AR overlays that provide step-by-step guidance on welding techniques, acceptable angles, and movement patterns.                                               | 4            | 6               | -             | 3          |
| <b>PC8.</b> Ensure accurate measurements during welding preparations and executions using AR tools on AR Based projections of weld joints, shapes & sizes for accurate placement. | 2            | 4               | -             | 2          |
| <i>Perform post-AR Guided welding operations</i>                                                                                                                                  | <b>10</b>    | <b>20</b>       | -             | <b>5</b>   |
| <b>PC9.</b> Allow welded joints to cool naturally or use appropriate cooling methods following the welding process to prevent warping or cracking.                                | 4            | 4               | -             | 1          |
| <b>PC10.</b> Utilize AR inspection tools overlay for visual inspection of welds.                                                                                                  | 4            | 8               | -             | 2          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                             | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC11.</b> Utilize AR sensors & cameras for providing feedback on weld quality, heat input & other critical parameters and other feedback. | 2            | 8               | -             | 2          |
| <b>NOS Total</b>                                                                                                                             | <b>30</b>    | <b>50</b>       | <b>-</b>      | <b>20</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                             |
|----------------------------|---------------------------------------------|
| <b>NOS Code</b>            | CSC/N0217                                   |
| <b>NOS Name</b>            | Perform AR-guided SMAW/GMAW in Shipbuilding |
| <b>Sector</b>              | Capital Goods                               |
| <b>Sub-Sector</b>          |                                             |
| <b>Occupation</b>          | Welding and Cutting                         |
| <b>NSQF Level</b>          | 4                                           |
| <b>Credits</b>             | 1                                           |
| <b>Version</b>             | 1.0                                         |
| <b>Last Reviewed Date</b>  | 08/05/2025                                  |
| <b>Next Review Date</b>    | 08/05/2028                                  |
| <b>NSQC Clearance Date</b> | 08/05/2025                                  |

## Assessment Guidelines and Assessment Weightage

### Assessment Guidelines

1. Criteria for assessment for the Qualification Pack will be created by CGSC.
2. Performance Criteria (PC) have been assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
3. The assessment for the theory part will/may be based on knowledge bank of questions approved CGSC.
4. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
5. Assessment Agencies will create Assessor Guides comprising of Theory and Practical Assessment Set and Guidelines for each examination/training centre (as per assessment criteria below). The same will be approved by CGSC for adequacy.
6. To successfully attain Certification on the Qualification Pack, the trainee must score a minimum of 70% in each Core NOS and minimum of 70% in all non-core NOS. In addition, a candidate needs to attain a minimum overall pass percentage of 70% for certification.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

## Qualification Pack

**Minimum Aggregate Passing % at QP Level : 70**

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

## Assessment Weightage

Compulsory NOS

| National Occupational Standards                                                                                          | Theory Marks | Practical Marks | Project Marks | Viva Marks | Total Marks | Weightage  |
|--------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|-------------|------------|
| CSC/N1335.Follow the health and safety practices at the work                                                             | 30           | 70              | -             | -          | 100         | 15         |
| CSC/N1336.Coordinate with co-workers to achieve work efficiency                                                          | 30           | 70              | -             | -          | 100         | 15         |
| CSC/N0204.Manually weld carbon and low alloy steels by using Metal Arc Welding (MMAW)/ Shielded Metal Arc Welding (SMAW) | 30           | 50              | -             | 20         | 100         | 20         |
| CSC/N0209.Manually weld metals by using MIG/MAG welding                                                                  | 30           | 50              | -             | 20         | 100         | 30         |
| DGT/VSQ/N0101.Employability Skills (30 Hours)                                                                            | 20           | 30              | -             | -          | 50          | 20         |
| <b>Total</b>                                                                                                             | <b>140</b>   | <b>270</b>      | <b>-</b>      | <b>40</b>  | <b>450</b>  | <b>100</b> |

Optional: 1 Perform AR-guided Oil & Gas Welding Scenario & Performance Evaluation

| National Occupational Standards                             | Theory Marks | Practical Marks | Project Marks | Viva Marks | Total Marks | Weightage |
|-------------------------------------------------------------|--------------|-----------------|---------------|------------|-------------|-----------|
| CSC/N0215.Perform AR-guided SMAW/GMAW in Oil & Gas Industry | 30           | 50              | 0             | 20         | 100         | 40        |
| <b>Total</b>                                                | <b>30</b>    | <b>50</b>       | <b>-</b>      | <b>20</b>  | <b>100</b>  | <b>40</b> |

## Qualification Pack

Optional: 2 Perform AR-guided SMAW/GMAW in Railways

| National Occupational Standards                   | Theory Marks | Practical Marks | Project Marks | Viva Marks | Total Marks | Weightage |
|---------------------------------------------------|--------------|-----------------|---------------|------------|-------------|-----------|
| CSC/N0216.Perform AR-guided SMAW/GMAW in Railways | 30           | 50              | 0             | 20         | 100         | 30        |
| <b>Total</b>                                      | <b>30</b>    | <b>50</b>       | <b>-</b>      | <b>20</b>  | <b>100</b>  | <b>30</b> |

Optional: 3 Perform AR-guided SMAW/GMAW in Shipbuilding

| National Occupational Standards                       | Theory Marks | Practical Marks | Project Marks | Viva Marks | Total Marks | Weightage |
|-------------------------------------------------------|--------------|-----------------|---------------|------------|-------------|-----------|
| CSC/N0217.Perform AR-guided SMAW/GMAW in Shipbuilding | 30           | 50              | 0             | 20         | 100         | 30        |
| <b>Total</b>                                          | <b>30</b>    | <b>50</b>       | <b>-</b>      | <b>20</b>  | <b>100</b>  | <b>30</b> |

## Qualification Pack

### Acronyms

|             |                                                 |
|-------------|-------------------------------------------------|
| <b>NOS</b>  | National Occupational Standard(s)               |
| <b>NSQF</b> | National Skills Qualifications Framework        |
| <b>QP</b>   | Qualifications Pack                             |
| <b>TVET</b> | Technical and Vocational Education and Training |

## Qualification Pack

### Glossary

|                                              |                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sector</b>                                | Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.                                                                      |
| <b>Sub-sector</b>                            | Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.                                                                                                                                                                                     |
| <b>Occupation</b>                            | Occupation is a set of job roles, which perform similar/ related set of functions in an industry.                                                                                                                                                                                                |
| <b>Job role</b>                              | Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.                                                                                                                                                                                |
| <b>Occupational Standards (OS)</b>           | OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts. |
| <b>Performance Criteria (PC)</b>             | Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.                                                                                                                                                                    |
| <b>National Occupational Standards (NOS)</b> | NOS are occupational standards which apply uniquely in the Indian context.                                                                                                                                                                                                                       |
| <b>Qualifications Pack (QP)</b>              | QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.                                                                                                                       |
| <b>Unit Code</b>                             | Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'                                                                                                                                                                                                        |
| <b>Unit Title</b>                            | Unit title gives a clear overall statement about what the incumbent should be able to do.                                                                                                                                                                                                        |
| <b>Description</b>                           | Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.                                                                                                                   |
| <b>Scope</b>                                 | Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.                                                                                            |



## Qualification Pack

|                                         |                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge and Understanding (KU)</b> | Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.                                                                                                                                               |
| <b>Organisational Context</b>           | Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.                                                                                                                                                                       |
| <b>Technical Knowledge</b>              | Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.                                                                                                                                                                                                                                                               |
| <b>Core Skills/ Generic Skills (GS)</b> | Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles. |
| <b>Electives</b>                        | Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.                                                                            |
| <b>Options</b>                          | Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.                                                                                                                                                          |